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ELECTRICAL COUPLER FOR RESISTIVELY HEATED REACTOR SYSTEMS

Abstract

An electrical coupler for a resistively heated reactor system including an electrical conductor having a first end and a second end, defining a thickness therebetween, each of the first end and the second end having a first interface in the first end and a second interface in the second end and, at least one opening extending from the first end to the second end, the at least one opening configured to allow a fluid to flow through the electrical coupler.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/758,642 filed 28-JUN-2024, which claims the benefit of U.S. Provisional Application No. 63/524,468 filed 30-JUN-2023, each of which is incorporated in its entirety by this reference.

TECHNICAL FIELD

[0002] This invention relates generally to the resistive heating field, and more specifically to a new and useful electrical coupler (e.g., electrode) and/or method of use in the resistive heating field.

BACKGROUND

[0003] Traditional catalytic reactor systems are heated via the external combustion of fuel. The heat of combustion is transferred into the process stream through the reactor wall by radiative and conductive heat transfer mechanisms to drive reactions inside the system. This allows the heat generation mechanism to be decoupled from the catalytic surface and process gas but suffers from poor heat utilization and generates large amounts of CO.sub.2 from the combustion process. Heat transfer efficiencies of 50% or less are typical of these systems. There is thus a need for more efficient and economic ways to heat reactor systems.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0004] A detailed description of various aspects, features, and embodiments of the subject matter described herein is provided with reference to the accompanying drawings, which are briefly described below. The drawings are illustrative and are not necessarily drawn to scale, with some components and features being exaggerated for clarity. The drawings illustrate various aspects and features of the present subject matter and may illustrate one or more embodiment(s) or example(s) of the present subject matter in whole or in part.

[0005] FIG. 1 is a perspective and planform representation of a schematic representation of an example of the electrical coupler.

[0006] FIG. 2A-2D are schematic representations of various embodiments of the electrical coupler for resistively heated reactor systems.

[0007] FIG. 3A is a schematic representation of an exemplary reactor utilizing an electrical coupler.

[0008] FIG. 3B is a schematic representation of an example of a mechanical interlock between an electrical coupler and a catalytic element.

[0009] FIG. 3C is a schematic representation of an exemplary interface between electrical couplers and a catalytic element.

[0010] FIG. 3D-3F are schematic representations of exemplary electrical coupler and catalytic element interfaces.

[0011] FIG. 4A-4G are representations of several exemplary contact types of electrical couplers.

[0012] FIG. 5A-5E are several views of schematic representations of an exemplary interface 500 for an electrical coupler (e.g., section perspective views, isometric views, etc.).

[0013] FIG. 6 is a schematic representation of an electrical coupler for resistively heated reactor systems.

[0014] FIG. 7 is an example of a radial coupler for a resistively heated reactor systems.

[0015] FIG. 8 is a schematic representation an example of a reactive component of a resistively heated reactor.

[0016] FIGS. 9A-9F are schematic representations of examples of reactive components for a resistively heated reactor with exemplary fluid flow paths represented.

[0017] FIG. 10 is a schematic representation of an example of a catalytic element integrated into a reactor.

[0018] FIG. 11 is a schematic representation of an example of an exemplary catalytic element with electrical connections across a width (e.g., diameter) of the catalytic element.

DETAILED DESCRIPTION

[0019] The following description of the embodiments of the invention is not intended to limit the invention to these embodiments, but rather to enable any person skilled in the art to make and use this invention.

1. Overview

[0020] As shown in FIG. 8, the system preferably includes an electrode pair (e.g., anode and cathode, electric couplers) and a catalytic element. The electrode pair and catalytic element are preferably configured to allow fluid (e.g., gas, liquid) flow through the electrode pair and the catalytic element. In preferred embodiments, the electrode pair are configured to have less pressure drop for fluid passing through than the catalytic element.

[0021] A specific example of the system can include an electrical coupler (e.g., for a resistively heated reactor system), the electrical coupler including an electrical conductor having a first end and a second end, defining a thickness therebetween, each of the first end and the second end having a first interface in the first end and a second interface in the second end, and at least one opening extending from the first end to the second end, the at least one opening configured to allow a fluid to flow through the electrical coupler.

[0022] In another specific example of the system, a system (e.g., a reactor, resistively heated reactor, etc.) can include a pressure vessel having a first vessel end and a second vessel end with a sidewall circumscribing each and extending therebetween, a first electrical coupler disposed at the first vessel end, the first electrical coupler having a first electrical conductor having a first end and a second end, defining a thickness therebetween, and coupled to the first circular end of the pressure vessel at least one opening extending from the first end to the second end, the at least one opening configured to allow a fluid to flow therethrough into the pressure vessel, a second electrical coupler disposed at the second vessel end, the second electrical coupler having a second electrical conductor having a third end and a fourth end, defining a thickness therebetween, and coupled to the second vessel end of the pressure vessel at least one opening extending from the third end to the fourth end, the at least one opening configured to allow a fluid to flow from the pressure vessel through the second electrical coupler, a catalytic element disposed within the pressure vessel, the catalytic element in contact with the first electrical coupler and the second electrical coupler.

[0023] In a third specific example of the system, a resistively heated reactor can include a proximal end (e.g., having a planar porous fluid inlet), at least one catalytic element having a first end and a second end with a sidewall extending therebetween, a first electrical coupler having a first end and a second end, defining an electrical coupler sidewall therebetween, the electrical coupler sidewall configured to circumscribe a first portion of the sidewall of the at least one catalytic element, a second electrical coupler having a first end and a second end, defining an electrical coupler sidewall therebetween, the electrical coupler sidewall configured to circumscribe a second portion of the sidewall of the at least one catalytic element, the second electrical coupler laterally spaced from the first electrical coupler, a first separator disposed between the first and second electrical couplers circumscribing the at least one catalytic element, a second separator disposed between the first and second electrical couplers circumscribing the at least one catalytic element, the second

separator spaced oppositely from the first separator and a distal end having a planar porous fluid outlet. In variations of the third specific example (as shown for instance in FIG. 11), the electrical coupler connections can be made across a diameter or width of the catalytic element (e.g., a first electrical coupler can be encircled, surrounded, enclosed, etc. within a catalytic element which can be encircled, surrounded, enclosed, etc. by a second electrical coupler).

[0024] In some variations of the first, second, or third illustrative examples above, the positive and negative couplers of individual assemblies (coupler-catalytic element-coupler assemblies) can be connected in parallel, series, or a combination of the two within a single pressure vessel (e.g., to enable a multi-cellular arrangement). These multi-cellular arrangements can be beneficial for tuning the voltage and/or current requirements of the reactor system.

[0025] However, other suitable reactor configurations, catalytic elements, and/or electrical couplers can be used.

2. Technical Advantages

[0026] Variants of the technology can confer one or more advantages over conventional technologies.

[0027] Thermal reactor systems often operate at high temperatures (e.g., exceeding 1000° C.) and pressures (e.g., exceeding 1 Bar) under corrosive gas (e.g., in the presence of one or more of carbon monoxide, hydrogen, oxygen, ozone, ammonia, carbon dioxide, sulfur oxides, nitrogen oxides, fluorine, chlorine, bromine, iodine, hydrogen sulphide, ethylene oxide, carbonyl sulphide, methyl mercaptan, ethyl mercaptan, propyl mercaptan, butyl mercaptan, hydrogen cyanide, silane, phosphine, etc.) conditions. Thus, variants of the invention can include a design for electrical couplers (e.g., contactors, electrodes, etc.) that can be used to provide electrical energy or current to a catalyst without significantly affecting reactor performance (e.g., without shortening a reactor lifetime). For example, structured electrical contacts can be designed to possess low resistivity relative to the catalyst (e.g., catalytic element), are fluidically permeable to fluid flow, and are stable to degradation (e.g., caused by corrosive gas environments, elevated pressures, elevated temperatures, etc.).

[0028] Second, variants of the electrical couplers can contain features that allow them to be used as structural members, electrical interfaces, pressure vessel components, and/or compliant sub-components (among others), thereby reducing a complexity of the reactor design and/or reducing a number of parts.

[0029] Third, variants of the technology can be designed to mitigate a risk of damage to an electrode. In variants that include a straight catalytic member where fluid flows substantially parallel to a long axis of the catalytic member, the downstream electrode is often exposed to the most reactive conditions (as shown for example in FIG. 9A) and thus can be prone to degradation. To overcome this problem, the catalytic member can be designed such that both (or all when more than two electrodes are used) electrodes are upstream of the catalytic member relative the fluid flow. For instance, the catalytic member can have a chevron (e.g., V-shape), a U-shaped, convex, arched, boustrophedonic, zigzag, serpentine, or other similar shape where the ends of catalytic member are on a common side (e.g., where electrode broad faces are connected to the catalytic member substantially normally as shown for instance in FIG. 9D). In other examples, the second electrode can be arranged orthogonal to the first electrode where fluid does not substantially flow through the second electrode.

[0030] However, further advantages can be provided by the system and method disclosed herein.

3. Thermal Reactor

[0031] The thermal reactor preferably functions to perform (e.g., facilitate, enable, initiate, maintain, etc.) a reaction between two or more chemical species. The chemical species are typically in the fluid phase (e.g., gas, liquid, plasma, etc.). However, one or more chemical species could be solid phase and/or other suitable phase(s) of matter. For example, the thermal reactor can be used to perform a reverse gas water shift reaction (e.g., $\text{H}_2 + \text{CO}_2 \rightarrow \text{H}_2\text{O} + \text{CO}$).

Additionally or alternatively, the thermal reactor can be used to perform steam methane reforming ($\text{CH}_4 + \text{H}_2\text{O} \leftarrow \text{CO} + 3\text{H}_2$), dry methane reforming ($\text{CH}_4 + \text{CO}_2 \leftarrow 2\text{CO} + 2\text{H}_2$), Haber process ($\text{N}_2 + 3\text{H}_2 \leftarrow 2\text{NH}_3$), Kvaerner process ($\text{C}_n\text{H}_m \leftarrow n\text{C} + m/2\text{H}_2$), and/or other suitable industrial processes (particularly using gas phase reactants and ideally gas phase products, endothermic reactions, etc.).

[0032] The thermal reactor often operates at standard pressures or greater (e.g., between 1 atm and 100 atm). However (e.g., to control a product ratio, to control a byproduct recovery, etc.), the thermal reactor could be operated under reduced pressure (e.g., vacuum, pressure less than 1 atm, etc.).

[0033] As shown for example in FIG. 8, the thermal reactor can include a reaction module (e.g., an electrical coupler or electrode and a catalytic element). The thermal reactor can optionally include one or more of a pressure vessel, interface, compressive element, inlets and/or outlets (e.g., reagent and/or product ports), and/or other suitable elements.

[0034] In some variants, the thermal reactor can include a plurality of reaction modules. The plurality of reaction modules can be arranged in parallel (e.g., fluid flows through each reaction module contemporaneously, concurrently, simultaneously, etc.), in series (e.g., fluid from one reaction module to a subsequent reaction module), and/or in any suitable combination thereof (e.g., a plurality of parallel reaction modules where one or more of the plurality include a plurality of reaction modules in series, branching reaction modules, etc.).

[0035] The reaction module preferably functions as a site for chemical reactions to be performed. The reaction module (e.g., a catalytic element thereof) preferably achieves a high operating temperature (e.g., greater than about 500° C., 600° C., 750° C., 800° C., 900° C., 1000° C., 1100° C., 1250° C., 1300° C., 1500° C., 2000° C., 2500° C., etc.), which can be beneficial for driving an equilibrium of the chemical reaction to preferred products. However, the thermal reactor can operate in any suitable manner.

[0036] The reactor temperature is preferably achieved via resistive heating (e.g., Joule heating, Ohmic heating, etc.). However, the reactor temperature can otherwise be achieved (e.g., dielectric heating, induction heating, etc. for suitable catalytic elements, reaction modules, etc.).

[0037] Energy to achieve the reactor temperature is preferably provided via renewable energy (e.g., in the form of electricity). When Joule heating (resistive heat) is used to power the system, it is preferred that the heat generation is closely coupled to the catalyst material to maximize heat transfer to catalytic sites. This format is characterized in that electrical feedthroughs are included that penetrate the reactor wall and supply large amounts of electrical power to catalytic elements housed within the reactor.

[0038] The electrical couplers (e.g., electrode) preferably function to provide electricity (e.g., electrical current, electrical energy, etc.) to the catalytic module. The electrical couplers preferably have a lower electrical resistance than the catalytic module (e.g., so that resistive heating occurs at the catalytic module rather than the electrical coupler).

[0039] Typically, two electrical couplers are used for each catalytic module (e.g., one acting as an anode and one acting as a cathode). However, greater numbers of electrical couplers can be used (e.g., a plurality of anodes and/or cathodes) which can be beneficial for improving a uniformity of heating of the catalytic element (e.g., by improving a uniformity of electric current, potential drop, etc. through the catalytic element). In some variants, a single electrical coupler can be used (e.g., where the electrical coupler is designed to have an anode and cathode acting regions and the catalytic element is designed to pass current throughout the catalytic element rather than shorting across a region of the catalytic element proximal the electric coupler).

[0040] In some variants, the electrical coupler can be designed to allow flow of fluid through the electrical coupler (e.g., as shown for example in FIG. 9A, FIG. 9C, FIG. 9E, FIG. 9F). In these variants, the electrical coupler preferably has a smaller pressure drop than the catalytic element

(e.g., the electrical coupler has a smaller resistance to the flow of fluid). In addition to or alternative to the pressure drop, the electric coupler can additionally or alternatively be designed to achieve uniform electrical properties (e.g., electric current, potential drop, etc.) within the catalytic element (typically achieved by having greater coverage or contact between the catalytic element and the electrical coupler), pass a target current (e.g., 0.005 A, 0.01 A, 0.05 A, 0.1 A, 0.5 A, 1 A, 1.5 A, 2 A, 5 A, 10 A, 15 A, 20 A, 50 A, 500 A, 5000 A, etc.), to mitigate a risk of reactions or degradation occurring on the electrical coupler (e.g., generally achieved by having a smaller exposed surface area), and/or can otherwise be designed. Benefits of these variants can include, the electrical coupler acting to mix or improve mixing of the fluid prior to entering the catalytic element, increase a residence time of the fluid within the catalytic element, and/or can provide other suitable benefit(s).

[0041] Examples of electrical coupler structures that can be used to achieve fluid flow therethrough include boreholes or through-holes in the electrical coupler, mesh electric coupler, fibrous electrical coupler, porous electrical coupler, and/or other such techniques for enabling a fluid to pass through the electrical coupler.

[0042] In variants that use engineered fluid flow structures (e.g., boreholes, through-holes, etc.), the flow structures for electrical coupler pairs can be aligned (e.g., such that absent an object between the electrical couplers the fluid can flow along a straight or linear path between openings), anti-aligned (e.g., such that absent an object between the electrical couplers the fluid can not flow along a straight or linear path between openings), randomly or imperfectly aligned (e.g., such that absent an object between the electrical couplers the fluid can flow along a straight or linear path between a portion of openings or a portion of the openings), and/or can have any suitable alignment.

[0043] Typically, the electrical coupler is arranged so that a surface normal of a broad face of the electrical coupler is parallel to a fluid flow direction (as shown for example in FIG. 9A, FIG. 9E, or FIG. 9F). However, the electrical coupler can be arranged so that a surface normal of a broad face of the electrical coupler is perpendicular to the fluid flow direction (as shown for example in FIG. 9B, FIG. 9D, or FIG. 10) and/or intersects the fluid flow direction at an arbitrary angle (as shown for example in FIG. 9C).

[0044] In some variations (as shown for example in FIG. 9E), the reaction module can be designed so that the electrical couplers are arranged upstream relative to the catalytic element within the fluid stream. These variations can be achieved, for instance, by using U-shaped (e.g., horseshoe shaped, keyhole, omega-shaped, moon-like, lecotropal, etc.), V-shaped (e.g., chevron), boustrophedonic, zigzag, sinusoidal, serpentine, arcuate, and/or other suitably shaped catalytic elements where the ends of the catalytic element are on the same side (e.g., a surface normal to each end is parallel). These variations can provide a technical advantage of maintaining the electrical coupler are a lower temperature region within the fluid flow (thereby reducing degradation of the electrical coupler). Additionally, or alternatively, the reaction module can be designed such that a single electrical coupler is in the fluid path (e.g., a surface normal to each end of the catalytic element is perpendicular), where the single electrical coupler in the fluid path is preferably upstream relative to the catalytic element. However, the electrical couplers can be arranged downstream relative to the catalytic element in the fluid flow (as shown for example in FIG. 9F) and/or one electric coupler can be arranged upstream of the catalytic element and another electric coupler can be arranged downstream of the catalytic element (as shown for example in FIG. 9A or FIG. 9C).

[0045] The design of the electrical coupler preferably permits high volumes of a fluid, such as a gas or liquid, to flow through the electrical coupler (contactor) and be directed through the catalytic element. The fluid flow path exiting the electrical coupler preferably does not adversely affect the interaction of the gas phase with the catalytic element. The electrical coupler preferably distributes the fluid flow evenly across the cross-section of the catalytic element. The gas pathway of the electrical coupler may be a single opening (e.g., a hole or gap 0.5 to 10 cm in diameter), a plurality

of smaller holes (0.01 to 5 cm in diameter), a tortuous pathway (e.g., porous, channels, gaps between fibers, etc.), and/or can otherwise be designed. In variants that use them, a plurality of smaller holes can allow for increased flow distribution across the catalytic element and provide a more distributed contact for current flow from the electrical coupler to the catalytic element (which can be beneficial for evenly heating the catalytic element). The pressure drop across the electrical coupler should be small compared to (e.g., <50% of, <30% of, <25% of, <20% of, <10% of, <5% of, <1% of, etc.) the pressure drop across the catalytic element (e.g., resistive element).

[0046] However, the electrical coupler can be designed such that fluid can only pass through a portion (or contact a portion of the surface of) of the electrical coupler (as shown for example in FIG. 9D) and/or fluid cannot pass through the electrical coupler (as shown for example in FIG. 9B). These variations can be beneficial for reducing a risk of damage to the electrical coupler resulting from passing heated fluid through the electrical coupler.

[0047] The electrical coupler is preferably formed from an electrically conductive material. The electrical coupler can be formed wholly or partially from electrically conductive material. In some variants, the electrical coupler can have an insulating material applied thereto, such as an applique or component fastened thereto.

[0048] The electrical coupler is preferably made of a refractory material (e.g., a semiconducting, electrically conductive, etc. refractory material), metal (e.g., refractory metal), and/or combination thereof (e.g., a cermet material). Preferred examples of electrical coupler materials include: carbon (e.g., graphite electrodes), silicon-carbide (e.g., silicon-silicon carbide, carbon-silicon carbide), tungsten carbide, molybdenum carbide, titanium carbide, vanadium carbide, chromium carbide, zirconium carbide, niobium carbide, molybdenum carbide, ruthenium carbide, rhodium carbide, hafnium carbide, tantalum carbide, tungsten carbide, rhenium carbide, osmium carbide, iridium carbide, platinum silicide, titanium silicide, vanadium silicide, chromium silicide, zirconium silicide, niobium silicide, molybdenum silicide, ruthenium silicide, rhodium silicide, hafnium silicide, tantalum silicide, tungsten silicide, rhenium silicide, osmium silicide, iridium silicide, neptunium silicide, titanium, vanadium, chromium, zirconium, niobium, molybdenum, ruthenium, rhodium, hafnium, tantalum, tungsten, rhenium, osmium, iridium, alloys or composites therebetween, and/or other similar materials. In some variations, the electrical coupler material can include dopants (e.g., metal dopants, p-type dopants, n-type dopants, etc.) to increase a conductivity of the electrical coupler.

[0049] The electrical coupler can be formed from a single continuous component, for example, as a continuous block of metal that is machined, molded, cast or otherwise shaped to possess the geometric features described herein. In various embodiments, electrical coupler can be formed from a single continuous component formed from any material as described herein. Electrical coupler can be configured to conduct electricity from one or more power interfaces due to the geometry and material of the device as well as allow fluid flow therethrough (e.g., by geometric considerations). The electrical coupler can additionally or alternatively be configured to couple to one or more external systems (e.g., inside or outside the reactor system) by nature of geometric features such as interfaces.

[0050] The catalytic element functions to drive (e.g., increase a reaction rate, decrease a reaction energy barrier, etc.) target chemical reactions. The catalytic element is preferably disposed between at least two electrical couplers (e.g., an anode and a cathode). However, a single electrical coupler (that acts as both cathode and anode) can be used (e.g., when an electrical path through the catalytic element passes through the majority of the volume of the catalytic element before connecting the anode to the cathode).

[0051] The catalytic element can be cylindrical, prismatic (e.g., pyramidal, prismatic, antiprismatic, parallelepipedal, cupolaed, frustral, cube, rectangular prism, triangular prism, pentagonal prism, etc.), toroid (e.g., square toroid, rectangular toroid, hexagonal toroid, degenerate toroid, etc.), conical, and/or can have other suitable shapes (typically but not necessarily a 3D

shape with two planar broad surface ends). Often the ends are antiparallel (e.g., opposing one another across an axis of the catalytic element). However, the ends can be orthogonal, parallel (e.g., U-shaped, V-shaped, chevron, boustrophedonic, zigzag, sinusoidal, serpentine, etc. with the ends in the same plane or otherwise having surface normal directed in the same direction), and/or can otherwise be related or have suitable structure. In some variants (as shown for example in FIG. 9D), electrical couplers can be configured to be bonded to the catalytic elements from the side rather than the ends.

[0052] The catalytic element can be partially formed from an electrically conductive or semiconducting material. In some variants, catalytic element can be formed wholly from electrically conductive or semiconducting material. In some variants, the catalytic element can be formed from an electrically conductive material with a catalytic material applied thereto.

[0053] The catalytic element can include: a substrate, a catalyst, and/or any suitable components. In some variants, the substrate can be formed from catalyst material (e.g., the substrate and catalyst can be the same).

[0054] The substrate functions to support the catalyst and/or heat the catalyst and/or fluid. The substrate can additionally or alternatively function to mix the fluid (e.g., by introducing turbulence into the fluid flow), increase a residence time of the fluid proximal the catalyst (e.g., by forming a tortuous pathway through the substrate), and/or can otherwise function.

[0055] The substrate is preferably formed from a refractory material, refractory metal, and/or combinations thereof (e.g., cermet). For instance, the substrate can be made from the same or a different material that the electrical coupler can be formed from (carbon, silicon-carbide, tungsten carbide, molybdenum carbide, titanium carbide, vanadium carbide, chromium carbide, zirconium carbide, niobium carbide, molybdenum carbide, ruthenium carbide, rhodium carbide, hafnium carbide, tantalum carbide, tungsten carbide, rhenium carbide, osmium carbide, iridium carbide, platinum silicide, titanium silicide, vanadium silicide, chromium silicide, zirconium silicide, niobium silicide, molybdenum silicide, ruthenium silicide, rhodium silicide, hafnium silicide, tantalum silicide, tungsten silicide, rhenium silicide, osmium silicide, iridium silicide, neptunium silicide, titanium, vanadium, chromium, zirconium, niobium, molybdenum, ruthenium, rhodium, hafnium, tantalum, tungsten, rhenium, osmium, iridium, alloys or composites therebetween, etc.). Other examples of substrate materials include oxides (e.g., titania, alumina, ceria, zinc oxide, zirconia, silica, etc.), nitrides, and/or other suitable materials (e.g., which may include dopants to modify the electrical properties of the substrate to promote uniform thermal and/or electrical distributions throughout the catalytic element).

[0056] The substrate preferably has a high specific surface area (e.g., to promote reaction sites) such as $\geq 1 \text{ m}^2/\text{g}$, $\geq 10 \text{ m}^2/\text{g}$, $\geq 100 \text{ m}^2/\text{g}$, $\geq 1000 \text{ m}^2/\text{g}$, and/or other surface area (e.g., BET surface area). However, the substrate can have any suitable specific surface area.

[0057] The substrate is preferably a porous material (e.g., a material with a porosity greater than 5%, a material with a solid material volume of at most 95%, etc.). For example, the substrate can be a foam (e.g., open-celled foam, stochastic foam, regular foam, etc.), a woven-fiber, periodic open cell structure, lattice, and/or can have any suitable structure. However, the substrate can additionally or alternatively be a solid material (e.g., mesh, ribbon, etc.), include engineered structures (e.g., a roughened surface to facilitate a large surface area), and/or can otherwise be formed.

[0058] The catalyst functions to reduce an activation energy of a target chemical reaction (e.g., increase a reaction rate, promote reaction centers, etc.). The catalyst is preferably disposed on (e.g., coated on, adhered to, absorbed on, adsorbed on, etc.) the substrate. Additionally or alternatively, the catalyst can be integrated into the substrate (e.g., where at least a portion of the catalyst material protrudes from the substrate material, can be disposed on the electrical coupler (e.g., to increase the total catalyst loading within the reaction module and/or reactor), and/or can otherwise be arranged.

[0059] Examples of catalyst materials (e.g., for the RWGS reaction) can include: oxides (e.g., iron

oxide, chromium oxide, copper oxide, aluminium oxide, zinc oxide, cerium oxide, iron oxide, manganese oxide, indium oxide, nickel oxide, spinel oxides, solid solution oxides, perovskite-type oxides, composites or combinations thereof, etc.), metal catalysts (e.g., platinum, palladium, gold, rhodium, ruthenium, copper, nickel, rhenium, cobalt, iron, molybdenum, etc.), phosphides (e.g., copper phosphide, nickel phosphide, tungsten phosphide, cobalt phosphide, molybdenum phosphide, combinations thereof, etc.), promoters (e.g., alkali metals, precious metals, bimetallics, etc.) and/or other suitable catalyst materials (e.g., for a specific reaction, combined with a specific substrate material, etc.).

[0060] The catalytic element and the electrical couplers can be connected mechanically (e.g., physically compressed together such as using a spring, pneumatic systems, hydraulic system, vacuum, motors, Belleville washer, etc.), chemically (e.g., using an adhesive such as carbon), and/or using any suitable connection mechanism.

[0061] The electrical coupler is preferably bonded to the catalytic element in a way that minimizes the electrical resistance (e.g., minimizes contact resistance or prevents high resistant phases from forming) and/or allows for the stable delivery of high current densities to the catalytic element. These bonding properties can provide a technical advantage as a high local resistance at the contact or contact interface can create hot spots and/or areas prone to degradation under reaction conditions. Local resistance can be minimized by using appropriate geometric design, materials selection, mechanical compression, chemical bonding, and other such processes that ensure intimate electrical and physical contact between the electrical coupler and the catalytic element (e.g., without adding stress such as during thermal expansion and contraction).

[0062] One exemplary method of bonding the catalytic element and the electrical coupler(s) includes using an interfacial adhesion layer applied to the ceramic substrate. For example, an adhesion layer is deposited on the SiC by any one of placement of foil at the interface, thermal spray, electroplating or chemical vapor deposition (CVD)/physical vapor deposition (PVD) with masking of remainder. Then any one of the following may be used to activate the adhesion layer to bond the catalytic element and the electrical couplers: induction heating, resistive heating, laser heating, and ultrasonic vibration.

[0063] Another exemplary method of bonding the catalytic element and the electrical couplers includes direct bonding through chemical bonding. For example, diffusion bonding can be utilized. By physically placing components proximate to one another and applying a force (e.g., exceeding 100 MPa), followed by heating (e.g., up to or above about 2000° C.), a chemical bond can be induced between the two materials. Additionally, or alternatively, a precursor paste can be used by placing a mixture of Si, C and a binding agent at the interface followed by heating to elevated temperatures using any of the heating methods herein (induction, resistive, radiative, laser, vibration, etc.) to form a chemical bond.

[0064] Another exemplary method of bonding the catalytic element and the electrical couplers can include coating (e.g., with SiC) via CVD, liquid phase SiC, and/or other suitable approach.

[0065] Another exemplary method of bonding the catalytic element and the electrical couplers can include manufacturing (e.g., using additive manufacturing) the electrical coupler and element in one operation. Post-processing (e.g. silicon infiltration or other treatment) can then optionally be used (e.g., to further modify the relative conductivities of the couplers and the catalytic element).

[0066] Additionally, or alternatively, any component described in the entirety of the disclosure herein may be mechanically coupled via a wetted connection, brazing, diffusion bonding, ultrasonic welding or the like.

4. Specific Examples

[0067] The methods and systems presented herein may be used for an electrical coupler. The disclosed subject matter is particularly suited for an electrical coupler for a resistively heated reactor system. For the purpose of explanation and illustration, and not limitation, a first specific example of a system in accordance with the disclosed subject matter is shown in FIG. 1 and is

designated generally by reference character **100**. Similar reference numerals (differentiated by the leading numeral) may be provided among the various views and Figures presented herein to denote functionally corresponding, but not necessarily identical structures. However, reference numerals with different leading numeral can all refer to the same structure (e.g., from different perspectives, different variations of the same structure, etc.)

[0068] The first specific example can be used for an electrical coupler. The disclosed subject matter is particularly suited for an electrical coupler for a resistively heated reactor system. For the purpose of explanation and illustration, and not limitation, an exemplary embodiment of the system in accordance with the disclosed subject matter is shown in FIG. **1** and is designated generally by reference character **100**. Similar reference numerals (differentiated by the leading numeral) may be provided among the various views and Figures presented herein to denote functionally corresponding, but not necessarily identical structures.

[0069] FIG. **1** is a perspective and planform view of a schematic representation of the first specific example of the electrical coupler **100** for resistively heated reactor systems in accordance with the disclosed subject matter. A plurality of electrical couplers **100** may be operationally, functionally and/or physically similar or identical, thus reference to electrical coupler **100** may be made to describe one or both of the electrical couplers, or in some figures, every electrical coupler present in a reactor or system of reactors. Electrical coupler **100** may include a first end **101** and a second end **102**, defining a thickness **103** therebetween. First and second ends **101**, **102** may be any cross-sectional shape. For example, and without limitation, the first and second ends may have a circular cross-sectional shape, thereby forming a cylindrical electrical coupler **100** with two planar, normal circular faces as termini on either side of thickness **103**. In various embodiments, first end **101** and second end **102** may have a rectilinear cross-sectional shape, oblong cross-sectional shape, or any polygonal cross-sectional shape. First end **101** and second end **102** may be parallel to one another, having a normal circular outer wall extending therebetween forming a right cylindrical electrical coupler **100**. For example, and without limitation, electrical coupler **100** may have non-parallel first and second ends **101**, **102**. For example, and without limitation, first end **101** and second end **102** may have surface features, such as ridges, ribs, rings, waves or other protrusions on the surface thereof. For example, and without limitation, the cylindrical wall extending between the first end **101** and the second end **102** may be angled, scalloped or include features such as bottlenecks, undulations, corrugations or the like. In various embodiments, first end **101** may be a different cross-sectional shape than second end **102**. In various embodiments, first end **101** may be the same or similar cross-sectional shape but a different size than second end **102**, thus forming a conical or conical sectional shape of electrical coupler **100**.

[0070] The first specific example of the electrical coupler **100** may have an interface **104** within the first end **101**. Interface **104** may be formed as one or more countersinks that have the same planform shape as the end in which it is sunk. For example, a circular end may have a circular interface **104** disposed therein. Interface **104** may have a countersink bottom of a similar or the same shape as interface **104**, for example, interface **104** may have a circular bottom. Interface **104** may include internal geometrical features such as ridges, ribs, or radial features such as shoulders and necks. Interface **104** may form a first interface with one or more sealing features therein. Similarly, second end **102** may have a second interface **105**, second interface **105** may be a mirror image of interface **104**. In various embodiments, the countersinks of interfaces **104**, **105** are different planform shapes, for example, first interface **104** may be a circular bore into the radial center of the first end **101** and second interface **105** may be a square bore into the radial center of the second end **102**. In various embodiments, each of the countersinks may be bored a different depth, for example, first interface **104** may be deeper than second interface **105**. Interface **104** and/or **105** may be an interface for sealing against a pressure vessel as described herein. In various embodiments, interface **104** and interface **105** may be concentrically formed in the same end of electrical coupler **100**.

[0071] For example, and without limitation, interfaces **104** and **105** may be radially centered with the electrical coupler **100** and have at least one opening **107** therethrough, thus connecting the first end **101** and the second end **102** or facilitating fluid flow therethrough. In various embodiments, the opening **107** may be radially centered and bored normally to electrical coupler **100**. The at least one opening **107** may be made at the bottom most portion of each of the interfaces **104** and **105**. In various embodiments, the opening **107** may be formed during the countersink fabrication process. In various embodiments, the at least one opening **107** is a borehole formed normally to the first end **101** and the second end **102**, the at least one opening **107** having a diameter of approximately 0.5 to 10 centimeters.

[0072] In variations of the first specific example, the at least one opening **107** is a plurality of openings formed from the first end to the second end, each of the plurality of openings having a diameter of approximately 0.01 to 1 centimeter (cm). In variations of the first specific example, the plurality of openings **107** are formed in a ring disposed concentrically with a normal axis extending from a center of the first end **101** to a center of the second end **102**. The mechanical design of the electrical coupler ideally allows for even fluid flow through the assembly and may need to distribute fluid evenly prior to entry into the catalytic heating element. When fluid is not evenly distributed across the electrical coupler prior to entry into the catalytic element, the efficiency of the element may be significantly reduced. In variations of the first specific example, the plurality of openings **107** may be formed normally to one or both of the first end **101** and the second end **102**. In variations of the first specific example, a first portion of the plurality of openings may be oriented normally with one or both of the first end **101** and the second end **102** and a second portion of the plurality of openings **107** may be formed at an angle thereto. In variations of the first specific example, electrical coupler **100** can be formed from a structure having a plurality of randomly aligned holes and channels, akin to foam, cork, or a sponge. In variations of the first specific example, at least one opening **107** is formed in the electrical coupler **100** as a plurality of channels and/or pores.

[0073] With continued reference to FIG. **1** and variations of the first specific example, electrical coupler **100** may have a power interface **108** formed therein. The power interface **108** may be formed or emplaced on one of the first end **101** (as shown) and extend through to the second end **102** (as shown). In various embodiments, the power interface **108** may be formed on the surface of the first end **101**, second end **102** or thickness **103**, exclusively. In variations of the first specific example, power interface **108** may be emplaced or formed integrally with the electrical coupler **100**. In variations of the first specific example, the power interface **108** may be configured to couple to one or more wires or conduits configured to provide electrical energy to the electrical coupler **100**. In variations of the first specific example, the voltage drop and heat generation from the electrical connection may be utilized as fluid flow pre-heat system (e.g., using a resistive material as an electrical contact may preheat the fluid flow prior to entering the pressure vessel for catalysis).

[0074] With continued reference to FIG. **1** and variations of the first specific example, electrical coupler **100** may have a mechanical interface **112**. Mechanical interface **112** may be formed or emplaced on one of the first end **101** (as shown) and extend through to the second end **102** (as shown). In variations of the first specific example, the mechanical interface **112** may be formed on the surface of the first end **101**, second end **102** or thickness **103**, exclusively. In variations of the first specific example, mechanical interface **112** may be emplaced, machined, bored, carved, cut or formed integrally with the electrical coupler **100**.

[0075] In variations of the first specific example, the mechanical interface **112** can be configured to accept a threaded rod therein. For example, and without limitation, a portion of mechanical interface **112** can be threaded such that a rod can be inserted with a portion exposed such that a nut, washer, pin or other retainment component can be threaded or coupled thereto. In variations of the first specific example, mechanical interface **112** may be configured to press fit with another

component, such as a rod. In variations of the first specific example, mechanical interface **112** can be disposed radially about a central axis of the electrical coupler **100** at regular intervals. In variations of the first specific example, mechanical interface **112** can include a plurality of mechanical interfaces **112** of the same or different type. For example, and without limitation, mechanical interface **112** can be one or more slots configured to retain a hook or other boss, thereby holding any component stationary relative to electrical coupler **100**. For instance, and without limitation, mechanical interface **112** may be adjustable or adjusted. For instance, the threaded rod may be rotated to tighten or loosen any two components relative to the electrical coupler **100**. For instance, mechanical interface **112** may facilitate a seal between two components by tightening via rotation of a threaded rod.

[0076] As a second specific example (that can be the same as and/or integrated with the first specific example and variations thereof with potential design modifications), the electrical coupler **100** is shown as a side view and planform view (e.g., as shown in FIG. 2). As can be seen in FIGS. 1 and 2A-2D, interface **105** can be an interface for one or more fluid supply lines, such as a fluid inlet or fluid outlet. In variations of the second specific example, interface **105** may include threads or another retainment feature configured to couple and seal a component carrying a fluid such as a gas to the electrical coupler **100** and provide gas through the at least one opening **107**. In variations of the second specific example, the electrical coupler can be fluidically and electrically conductive and stable at high temperature (e.g., $\geq 500^{\circ}\text{C.}$, $\geq 900^{\circ}\text{C.}$, etc.) in corrosive environments.

[0077] With continued reference to FIG. 2A and in variations of the second specific example, second interface **105** is shown in section view and with phantom line. The second interface **104** can form a second interface for a catalytic element (to be described herein below) and/or a pressure vessel. The second interface (**104**) can include any number of geometric or materials features such as internal shoulders and rubber gaskets to form a bond with electrical coupler **100** (and seal the connected).

[0078] Referring now to FIG. 2B-2D, variations of the electrical coupler of specific example 1 or 2 **100a** and **100b** are shown in isometric views. In various embodiments, electrical couplers **100** can be the same on either end of the catalytic element, secured thereto through the same or similar mechanism. In various embodiments, electrical coupler **100a** may configured separately than electrical coupler **100b**. For example, and without limitation, electrical coupler **100a** electrical can be configured to couple to the catalyst or heater element through a compressional set screw connection. However, other suitable fastening mechanisms can be used (e.g., springs). Electrical coupler **100** can have an opening **107** configured to allow for fluid flow therethrough. Opening **107** can be threaded and configured to receive a set screw, the set screw rotating to compress and form an electrical contact with the catalytic element. Additionally, or alternatively, electrical coupler **100b** can be configured to have an opening **107**, the opening **107** having a plurality of smaller openings disposed therein. The opening **107** in electrical coupler **100b** may be configured to circumscribe the catalytic element and form a mechanical bond therebetween.

[0079] In some variations of the electrical couplers **100a** and/or **100b** (as shown for example in FIG. 2C or FIG. 2D) can have an asymmetrical stand-off **204**. The stand-off **204** can have a threaded hole therein and formed parallel to the stand-off **204**. The (optionally threaded) hole can be configured to form an electrical connection with one or more power interfaces as described below, for example a wired power connection or the like. In various embodiments, stand-off **204** may be configured for electrical coupling in a plurality of methodologies, for example soldering or bolt-on to an electrical lead. The threaded stand-off **204** can be configured to be screwed in a power conductor, wherein the power conductor is sealed to the reactor vessel via a feedthrough, which separates the vessel from ambient atmosphere. In various embodiments electrical couplers **100a**, **100b** can be configured to be bonded to the catalytic element. The radial symmetry of the counter-sink and the backstop with a plurality of holes can be configured to match the geometry of any catalytic elements.

[0080] As a third specific example (e.g., of a reactor **300** such as shown for example in FIG. 3A), the reactor can utilize an electrical coupler **100** for resistively heated reactor systems. Reactor **300** can include an electrical coupler **100** (or substantially similar conductor, electrode, contactor, etc.) disposed at a distal end and a proximal end thereof. Reactor **300** can include a pressure vessel **304**. Pressure vessel **304** can have a first circular end and a second circular end, with a cylindrical wall circumscribed about the perimeter of each and extending therebetween to form a cylindrical vessel. However, the pressure vessel can have a prismatic shape, toroidal shape, and/or other suitable vessel structure can be used. The pressure vessel **304** can be seated within the countersinks (**105**) of each of the electrical couplers **100a**, **100b**. The pressure vessel **304** can be sealed against the electrical couplers **100a**, **100b** (e.g., via one or more O-rings, gaskets, knife-edge seals, etc.). In various embodiments, the pressure vessel **304** can form a press fit with one or both of the electrical couplers **100a**, **100b**. In various embodiments, the pressure vessel **304** can be configured to not allow ingress or egress of a fluid, such as gas or liquid flow **312**, through the walls thereof. The pressure vessel can be formed, for example, from quartz crystal and/or other suitable materials that are resilient to the reaction conditions within the reactor. The pressure vessel **304** can be coupled to one or both of the electrical couplers **100a**, **100b** via chemical bonding or adhesion, can be integrally formed with the one or both of the electrical couplers **100a**, **100b**, can be mechanically coupled to one or both of the electrical couplers **100a**, **100b**, and/or can otherwise be configured. [0081] In various embodiments, pressure vessel **304** can have a sealing feature at the first circular end and second circular end configured to matingly couple with a corresponding feature of the first and second electrical couplers **100a**, **100b**. For example, the rim of the pressure vessel **304** can seat within a circular channel formed within the countersink of the electrical coupler **100a** or **100b**. In various embodiments, the pressure vessel can be configured to accept a flowing fluid entering said vessel from the opening in the electrical coupler **100a** and conduct said fluid flow **312** through the second electrical coupler **100b** (e.g., via the catalytic element). In various embodiments, H.sub.2-stable and carburization-resistant materials can be utilized to create the weldment and any or all reactor seals.

[0082] In various embodiments, one or both of first electrical coupler **100a** and second electrical coupler **100b** can have a power interface (e.g., as described in reference to the first specific example). In various embodiments, one or both of first electrical coupler **100a** and second electrical coupler **100b** can have a mechanical interface or plurality thereof (e.g., as described in the first or second specific example). In various embodiments, each of the electrical couplers **100a**, **100b** can have corresponding mechanical interfaces such as fastener interfaces, configured to couple the reactor **300** together. For instance, at least one threaded rod can extend from the first electrical coupler **100a** to the second electrical coupler **100b**, parallel and exterior to the cylindrical wall of pressure vessel **304**, the threaded rod configured to compress and couple the reactor **300** components. In various embodiments, first electrical coupler **100a**, second electrical coupler **100b**, and pressure vessel **304** may be formed as concentrically and serially disposed cylinders. For example, pressure vessel **304** may have a smaller diameter than each of electrical couplers **100a**, **100b**. In various embodiments, pressure vessel **304** may have the same diameter as one or both of electrical couplers **100a**, **100b**. In other embodiments, pressure vessel **304** may have a larger diameter as one or both of electrical couplers **100a**, **100b** with the couplers and catalytic element contained within the pressure vessel.

[0083] In the third specific example, the reactor **300** can include a catalytic element **308**. Catalytic element **308** is sealably (e.g., hermetic seal, fluidic seal, etc.) disposed within the pressure vessel **304**, the catalytic element in electrical contact with the electrical couplers **100a**, **100b**. In various embodiments the catalytic element **308** is in direct electrical contact with each of the electrical couplers **100a**, **100b**. In various embodiments, the catalytic element is in indirect electrical contact with the electrical couplers **100a**, **100b**. The catalytic element **308** can radially contact (e.g., electrical coupler **100** can circumscribe the sides of the catalytic element **308** to allow for

unobstructed fluid flow **312**), axially contact (e.g., electrical coupler **100** can be attached to the ends of the catalytic element **308** to allow for fluid flow **312** through the couplers and catalytic element, where fluid flow can occur radially or orthogonally to the electrical couplers, etc.), and/or can other contact the electrical coupler **100a** or **100b**. In various embodiments, one or more components can optionally be included and configured to facilitate electrical communication between the electrical couplers and the catalytic element, such as compressive elements like washers, metal foams and the like. In various embodiments, the electrical couplers **100a**, **100b** have a matched thermal expansion coefficient to the catalytic element **308** and forms a low-resistance contact with the catalytic element. The resistance of the electrical couplers is preferably lower than that of the catalytic element **308** (so that the catalytic element is the primary site of heating), though this is not strictly necessary. As a specific embodiment, catalytic element may be 3-6 inches in wide and 6-12 inches long; however, other suitably dimensioned materials can be used.

[0084] In the third specific example, the reactor **300** can be arranged such that the electrical couplers **100a**, **100b** provide for unobstructed fluid (e.g., gas flow) flow **312** through to the catalytic element **308** within the pressure vessel **304**, to provide for current flow to resistively heat the cell, and to provide a sealing surface for the pressure vessel **304**. Reactor **300** can optionally include one or more thermocouple emplaced therein, each configured to measure the temperature in a localized or larger portion of reactor **300**. The pressure vessel **304** can be formed from one or more insulating materials configured to improve thermal efficiency of system while maintaining an optimal fluid flow **312** profile. In various embodiments, pressure vessel **304** may interlock with one or both electrical couplers **100a**, **100b** to form a seal against fluid flow **312**.

[0085] In a specific example (as shown for example in FIG. 3B), a mechanical interlocking bond between electrical coupler **100** (which may be one or both of electrical couplers **100a**, **100b**) and catalytic element **308** can be used. The electrical coupler **100** and catalytic element **308** can have corresponding geometric features configured to be brought into contact to form a mechanically interlocking bond (e.g., via twisting, via tight manufacturing tolerances, via mating structures, etc.). In various embodiments, the electrical coupler **100** can have a circular opening **107** in the axial center thereof, opening **107** having two linear bosses forming segments of the opening **107** defined by the linear chords. The segments can be formed from the same material as the electrical coupler **100**. In various embodiments, the segments can be formed from a different material than the electrical coupler **100**. In various embodiments, the segments can be integrally formed or affixed after fabrication of the electrical coupler **100**. Correspondingly, the catalytic element **308** can be formed with a prismatic end portion configured to enter the opening **107** between the segments. Then the catalytic element **308** can be rotated (or otherwise mated), capturing the segments of the electrical coupler **100** in matingly-formed cutouts in the prismatic end of the catalytic element **308**. The segments can be configured to retain the catalytic element **308** in the axial direction of the electrical coupler **100**. This mechanically interlocking technique is only illustrative and not intended to limit the types of mechanical bonding between components described herein.

[0086] A specific example of an interface between electrical couplers **100a**, **100b** and catalytic element **308** is shown in FIG. 3C. Electrical couplers **100a**, **100b** can be formed as generally planar components with an arcuate section that is sized to the end of catalytic element **308**. For example, catalytic element **308** can have a circular end shape, so electrical couplers **100a**, **100b** can have generally circular portion to abuts thereto. Electrical couplers **100a**, **100b** can have a tab portion that extends linearly from the center of the arcuate portion, the tab portion configured to receive one or more electrical connections, such as alligator clips, bolt-on coupling or the like. In various embodiments, the tab portion can be formed as a generally triangular section. In various embodiments, the tab portion can be any shape that extends away from the end of the catalytic element **308** to couple an electrical connection thereto. In various embodiments, tab portion can have power interface **108** emplaced thereon, for example as an opening sized to receive an electrical cable. In various embodiments, the tab can be a feature (rod-like, plate-like, or of any

other shape) that extends away from the catalytic element sufficiently to mate with external power conductors in a region of the system that is of significantly lower temperature (a “cold-zone”), such as to enable mating to more highly conductive power feedthroughs that are not able to withstand the catalytic conditions close to the element (a “hot-zone”).

[0087] In one specific example, catalytic element **308** may be formed as a rectangular prism (as shown for example in FIG. 3D), having an approximately square cross section. Machined carbon (or other suitable electrical coupler material) can form an electrode and be bonded to the side of the prismatic catalytic element (e.g., by conducted cement, which may also be formed from carbon). These electrodes can be bonded to the catalytic element **308** on opposite sides, sandwiching the element therebetween. Additionally, or alternatively, on the two other sides of the catalytic element, machined alumina may be bonded to the catalytic element **308**. The alumina may be utilized for its electrically insulative and thermally conductive properties. Additionally or alternatively, all sides (e.g., faces) of the catalytic element **308** can be bonded to the electrodes, a subset of sides (such as opposite sides) can be bonded using conductive carbon cement and the other pair of sides can be bonded using alumina cement or alumina paste, and/or other suitable binding mechanisms can be used on other suitable surfaces (e.g., sides).

[0088] In a specific example a catalytic element **308** may be formed from a foam substrate (as shown for example in FIG. 3E in perspective and planform views). Said foam substrate can be formed from one or more metals, refractory materials (e.g., ceramics), combinations thereof, and/or other suitable material(s). The foam catalytic element **308** can be subject to thermal spray coatings (e.g., nickel, Hastelloy, etc.) on at least one side. The thermal spray coating may be applied on any side of catalytic element **308**, including opposite and opposing sides, or any arrangement with electrical insulators.

[0089] In a specific example (as shown for instance pictorially in FIG. 3F), catalytic element-electrode assemblies (e.g., reaction modules) can include paste (e.g., carbon, silicon carbide, etc.) applied to the catalytic element **308** to bond tabs (e.g., nickel tabs) to opposite and opposing sides. The tabs may be formed in any shape or dimension, here shown as foil having a rectangular planform shape. Said catalytic element **308** can be formed as a rectangular prism having generally planar sides. On the opposing side, the catalytic element **308** can be bonded to graphite components using any of the paste/cements described herein. The graphite components may have integrally formed or affixed tabs for electrical connection via clamps, in various embodiments.

[0090] In these specific examples (e.g., the first second or third specific example) the electrical couplers **100a**, **100b** can have features that allow for the interaction with other external or internal systems as described above (as shown for example in FIGS. 4A-4F). These features can be used to interact with external power supply lines (such as power interface **108**), insulating components, fluid flow **312** components, mechanical interface **112** features (e.g., for the stability of system), and/or other such systems. These features can also provide sealing surfaces for pressure vessels **304**, insulating jackets, and/or other components. In various embodiments, one or more insulating jackets can be emplaced or coupled around the reactor **300**. In various embodiments the insulating jackets can be electrically and/or thermally insulating. In various embodiments, the insulating jacket can include structural components configured to couple, retain, or stabilize at least a portion of the reactor **300**. In various embodiments, the insulating jackets can be either internal or external to the pressure vessel and can act to electrically isolate various sub-components of the reactor system.

[0091] In a specific variation of the third specific example, the reactor **300** can be mechanically connected by threaded rods extending from electrical couplers **100a**, **100b** (as shown for example in elevation views in FIG. 4B). The electrical couplers can have corresponding holes where a straight rod can be seated within both couplers and extend therebetween, the couplers fastened to the rods. The electrical contacts can be any as described herein, for example a metallic foam, a washer, spring or the like. Reactors **300** can include gaskets configured to form a seal between the

electrical couplers **100** and the pressure vessel **304**. There can be a space (e.g., annular space) between the catalytic element **308** and the pressure vessel **304**, or insulation jacket. In various embodiments, the pressure vessel and/or insulation jacket can have thermocouple feedthroughs **402** disposed therein, configured to receive a thermocouple and allow for electrical coupling exterior to the reactor **300**. In other embodiments, the space may be filled by inert, insulating media to limit gas bypass around the catalytic element.

[0092] In some variations of examples of the reactor **400a** (as shown for instance in the elevation view of FIG. 4C) can utilize a bolted-on tab coupling between the electrical couplers **100a** and the catalytic element **308**. The tabs **404** can be formed from tungsten (or other refractory metals, refractory material, combinations thereof, etc.) and have an opening formed therein (e.g., configured to receive a bolt or mechanical fastener to form a bond to the electrical couplers **100a**, **100b**). The tabs **404** may have slots, holes or a combination thereof formed therein to receive a correspondingly shaped mechanical fastener therethrough. In various embodiments, the tabs **404** can be bonded to the catalytic element using any suitable bonding techniques. In various embodiments, the tabs **404** can be formed from any electrically conductive material. Reactor **400b** can have electrical couplers **100a**, **100b** mechanically or chemically (or both) bonded to the catalytic element **308** as described herein. The electrical coupler **100a**, **100b** can, for instance, be partially or wholly formed from graphite. In various embodiments, reactors **400a**, **400b** can have a graphite-based current-carrying thermal expansion joint **408** configured to maintain electrical connection between the electrical couplers and the catalytic element during thermal expansion. The thermal expansion joint **408** can include one or more wires and/or other electrical connections configured to move and maintain the electrical connection during movement of the components, such as electrical coupler **100a** and/or catalytic element expansion and/or contraction. In various embodiments, thermal expansion joint **408** can be configured to deliver electric current to the coupler **100a** via the discreet connections, the connections radially spaced about coupler **100a** to provide for even heating and even expansion of the coupler **100a**.

[0093] As shown for instance in elevation and isometric views in FIGS. 4E-4G, these specific examples of the reactor **400c** (which can be the same as any preceding reactor with a different number) can include an insulation jacket **412** disposed around the catalytic element **308** and the pressure vessel **304**. In various embodiments, the insulation jacket **412** can be formed partially or wholly by the pressure vessel **304**. In various embodiments, the insulation jacket **412** can be physically touching the pressure vessel **304**. In various embodiments, the insulation jacket **412** can be spaced from the pressure vessel **304** (e.g., forming an annular space therebetween). In various embodiments, the insulation jacket **412** can have the same planform shape as the pressure vessel **304**, for example circular. In various embodiments, the insulation jacket **412** can have a different planform shape, for example the pressure vessel can be a cylinder (with a circle cross section) and the insulation jacket **412** can be prismatic, having a rectangular cross section. Insulation jacket **412** can have any number of thermocouple feedthroughs **402** spaced along the axial direction, or radially about the perimeter. In various embodiments, insulation can alternatively or additionally be disposed within the pressure vessel. Additionally, the pressure vessel itself can be any material that is able to withstand the reaction conditions. For example, and without limitation, stainless steel, Inconel, refractory-lined pipe (e.g., using a pipe made from steel, stainless, etc.), or other suitable materials may be used.

[0094] As shown for instance in FIGS. 4F-4G, the exemplary reactors **400d**, **400e** (which can be the same as or distinct from other exemplary reactors) can have pressure vessels that are separated by an interstitial connection point along the axial direction of the reactor **400d**, **400e**. In various embodiments, pressure vessel **412** can be formed from any number of components and mechanically coupled together via one or more mechanical fasteners such as nuts and bolts threaded through corresponding flange holes. Reactor **400d** is depicted with both power interfaces **108** and fluid flow interface **312** formed in a first end, while reactor **400e** is depicted with the fluid

flow interface **312** disposed on both ends of the reactor. The pressure vessel **412** can be configured to have any number or type of interfaces to couple to various external systems (e.g., sensors, gas, power, etc.). In these exemplary reactors, insulation can be included between the pressure vessel walls and the catalytic element (and/or at other suitable locations within the reactor).

[0095] The specific examples as represented in FIGS. **1**, **3** and **4** show a schematic design of an electrical coupler (e.g., cylindrical conductor, contactor, electrode) **100** in which one side has a threaded connection for mating to a gas supply line, while the other side has a countersink design to accept a catalytic (resistive) element **308**. The catalytic element **308** can be bonded to the electrical coupler **100a**, **100b** using mechanical and/or chemical bonding techniques. Reactor **300** is shown with electrical couplers (contactors) **100a**, **100b** as described herein above. Additionally, reactor **300** is shown to have a fluid flow **312** flowing through a conduit coupled to the electrical coupler **100a**. One of ordinary skill in the art would appreciate that a threaded tube connection as shown is only one example of a system and method of delivering process gas to the reactor **300**.

[0096] Exemplary reactor **300** has a pressure vessel **304** sealably coupled to (e.g., forming a hermetic seal between) electrical couplers **100a**, **100b** and extending therebetween. Electrical couplers **100a**, **100b** are shown to be coupled together via a plurality of electrically isolated threaded rods extending therebetween, fixed in place by washers and nuts threaded on opposite and opposing surfaces of each. However, other coupling mechanisms can be used. Reactor **300** has catalytic element **308** disposed within pressure vessel **304**. The catalytic element **308** is shown as red and yellow-hot as it is heated by electrical energy provided by contact with one or both of electrical couplers **100a**, **100b**. One of ordinary skill in the art would appreciate that the depiction of the catalytic element **308**, regardless of pictorial coloration does not intend to limit it the temperatures of which the reactor **300** operates at, is capable of withstanding or optimized for. Additionally, one of ordinary skill in the art would appreciate that the pictorial representations herein do not intend to limit the types of reactions that may take place within reactor **300**.

[0097] The interface **500** can include the at least one opening **107** (e.g., as shown for example in a threaded configuration in FIGS. **5A**, **5C**, **5D**, and **5E**). In various embodiments (e.g., where the at least one opening **107** is formed as a plurality of openings), each opening of the plurality can be radially spaced about a central axis of the electrical coupler **100**, each opening can be threaded, or another mechanical interface can be utilized. In some variations, the interface **500** can include a set screw **504**. Set screw **504** can be configured to thread within the at least one opening **107**. The set screw **504** may be configured to enter the opening **107** from above (e.g., defined relative to FIG. **5**) and be rotated down the opening **107** until it is proximate the second side of the electrical coupler **100**, near the interface **105** (not shown). Set screw **504** can have a hexagonal opening formed therein, configured to mate with one or more standard tools, such as an Allen key, Allen wrench, Hex key, or the like. In various embodiments, set screw **504** can include geometric features configured to matingly-couple with any number of tools, such as Phillips or flathead screwdriver, drill driver, impact driver or the like.

[0098] In various embodiments, set screw **504** may be formed from the same material as electrical coupler **100**. In various embodiments, set screw **504** can be formed from any electrically conductive material, partially insulating material, semiconducting material, material combination (e.g., alloy, cermet, etc.), and/or other suitable material (e.g., electrical coupler material, substrate material, catalyst material, etc.). In various embodiments, set screw **504** can have a similar or the same coefficient of thermal expansion as the electrical coupler **100** (e.g., be formed from the same material).

[0099] In various embodiments, set screw **504** has one or more openings **506** configured to allow fluid flow therethrough. As shown for example in FIGS. **5A** and **5B**, set screw **504** can include a plurality of openings regularly spaced within the proximate side of the set screw to the catalytic element **308**. Additionally or alternatively, there may be a single opening **506**. In various embodiments, any of the openings **506** can be irregularly shaped, regularly shaped, circular,

oblong, slots, a mesh/screen, and/or have any suitable structure. In various embodiments, interface **500** can include a compressive element **508** (for example a washer). In various embodiments, the compressive element **508** can be compressed by the rotation of the set screw **504** and driven into contact with the catalytic element **308**. The compressive element **508** can be formed from any material (e.g., catalyst material, substrate material, electrical coupler material, interface material, etc.). Compressive element **508** can be configured to deform, whether elastically or plastically in response to the compression from the set screw **504**. For example, compressive element **508** can be a disc spring or Belleville washer, where the washer is formed with an angle surface, such that the outer and inner perimeters lie out of plane with one another. The set screw **504** can compress the perimeter of the washer spaced from the top of the catalytic element **308**, deforming the washer downward and ensuring contact between the electrical coupler **100** and the catalytic element **308** by the resistive spring force of the washer opposing compression. In another example, the compressive element **508** can be a metal foam. The metal foam can have a spring constant or shape memory, such that when compressed by the set screw **504** downward, the metal foam opposes said compression and presses back between the set screw **504** and the catalytic element **308**, thereby maintaining electrical communication therebetween. In various embodiments, compressive element **308** can be a spiral spring, leaf spring, internal or external tooth washer, helical washer, crinkle washer, split lock washer, fender washer, flat washer or the like.

[0100] As an illustrative example, the compressive element **508** can be a washer. The washer can be bonded to the catalytic element **308** and pressed against the electrical coupler **100**. In various embodiments, electricity may flow from the electrical coupler to the compressive element **508** to the catalytic element **308**, to a second compressive element **508** to the second electrical coupler **100**. In various embodiments, the compressive element **508** may be bonded with the catalytic element **308** via a carbon paste.

[0101] As shown for example in FIG. 6, a multi-cell reactor system (e.g., reactor **600**) can include a plurality of electrical couplers and catalytic elements (e.g., reaction modules). In this example of the reactor, the larger contacts at the ends can provide reactant gas delivery and product gas exit interfaces, while the intermediate electrical couplers can enable the sealing and electric isolation of adjacent cells with the use of a dielectric intermediate component (e.g., a quartz tube).

[0102] Exemplary reactor **600** may include one or more reactors **300** emplaced therein. For example, and without limitation, reactor **600** can include any number of cells, each cell formed by an individual reactor **300**. Reactor **600** can be a two-cell reactor system with multiple electrical couplers arranged to create a single fluid flow path through multiple resistive elements arranged in series with external power supply line interfaces. For instance, reactor **300a** can have an electrical coupler **100a** shown at the topmost portion with a pressure vessel **304** sealably coupled therewith, where the pressure vessel **304a** has a catalytic element **308a** therewithin. In examples of a multicell reactor, each cell can be in fluid communication by a common pressure vessel **304** and/or the reaction cells can be fluidically isolated. The pressure vessel **304** can extend from reactor **304a** to reactor **304b**, with electrical couplers **100b**, **100c** emplaced between the proximal and distal ends of the reactor **600** for internal sealing and inter-cell isolation. Additionally, or alternatively, each reactor **300a**, **300b** can have a separate pressure vessel in fluid communication with one another. For example, and without limitation, the first pressure vessel **304a** can extend from first electrical coupler **100a** to second electrical coupler **100b**, where each of the couplers contacts the catalytic element within the pressure vessel **304a**. Additionally or alternatively, a bridge pressure vessel **604** can extend between the second and third electrical couplers **100b**, **100c**. Then a second pressure vessel **304b** can extend from the third electrical coupler **100c** to the fourth electrical coupler **100d**, with the second catalytic element **308b** in electrical communication with both the third and fourth electrical couplers **100c**, **100d**. The catalytic elements **308a** and **308b** can extend within the pressure vessel between electrical couplers, for example catalytic element **308a** can extend from **100a** to **100b** and catalytic element **308b** can extend from **100c** to **100d**. In various embodiments,

each cell can have its own pressure vessel that are fluidly and/or electrically isolated from one another (e.g., each having their own fluid inlets). In various embodiments, each cell of reactor **600** can operate at a distinct temperature. In various embodiments, a common negative electrode may be used in the multi-cell system to improve volumetric efficiency. For example, two cells in series may use a shared contact as a common ground to minimize space in reactor **600**.

[0103] In various embodiments (as shown for an exemplary radial reactor **700** in FIG. 7 for instance), one or more power interfaces can be electrically connected to one or more of electrical couplers **704a**, **704b**. In various embodiments, electrical couplers **704a** can be a positive electrode and electrical couplers **704b** can be negative electrodes. In various embodiments, each electrical couplers **704a**, **704b** can each be electrically connected to a separate power source. In various embodiments electrical couplers **704a** can each be electrically connected to one another, wherein current flows through each of the electrical couplers **704a**. In various embodiments, electrical couplers **704a** can have varying resistance or material properties configured to evenly heat the radial reactor **700** from a single power interface. For example, and without limitation, the end electrical coupler **704a** can have a lower resistance than the middle electrical coupler **704a**, such that each of the electrical couplers provide even heat to the catalytic elements. In various embodiments, each of the radial electrical couplers **704a**, **704b** can be configured such that current flows radially across the elements. The power interfaces can be formed as feedthroughs configured to electrically connect to one or more patterned fingers.

[0104] In various embodiments, radial reactor **700** can include one or more longitudinal separators **708**. The separators **708** can be formed from one or more ceramics (and/or other suitable electrically insulating materials). In various embodiments, separators **708** can have the same radius as each of the electrical couplers **704a**, **704b** and electrically isolate electrical couplers **704a**, **704b** from one another. Separators **708** can extend from a distal end of the radial electrical coupler **700** to a proximal end of the radial electrical coupler **700**, providing both structural rigidity, electrical isolation, and fluid flow directionality. In various embodiments, fluid flows through a porous inlet **712** through the longitudinal axis of radial electrical coupler **700**. Separators **708** can be configured to direct the fluid flow longitudinally through the central portion of the reactor **700**.

[0105] Examples of radial reactor **700** can include a plurality of ring-shaped elements **716**, each associated with a electrical coupler **704a**, **704b**. In various embodiments, each electrical coupler **704a**, **704b**, and/or a subset thereof can have electrical coupling interfaces for electrical connection to one or more wires or other means of conveying electrical current. In various embodiments, the electrical coupling interfaces can include tabs, holes, clamps, or the like. In various embodiments, the catalytic elements **716** can be the same or similar to catalytic element **308** (and/or other catalytic elements described herein). In various embodiments, proximate elements can be of the same or different materials. In various embodiments, one or more components such as a current distributor can be connected to each of the catalytic elements **716** such that the patterned fingers will physically abut and compress. In various embodiments, each of the catalytic elements **716** can be electrically and/or fluidically connected in series. In various embodiments, each the catalytic elements **716** can be electrically and/or fluidically connected in parallel. In various embodiments, a first portion of catalytic elements **716** can be connected in series and a second portion can be connected in parallel. In various embodiments, fluid can flow around the outside of the radial coupler **700** and/or through the catalytic elements **716** stacked to form it. In various embodiments, each catalytic element **716** in the stack can be heated by its own corresponding electrical couplers **704a**, **704b**. In various embodiments, each catalytic element **716** can be selectively heated by its electrical coupler **704a**, **704b** at a different rate than adjacent or other catalytic elements **716**. In various embodiments, each electrical coupler **704a**, **704b** can operate independently of the others. In various embodiments, each catalytic element **716** can be sized or configured to be evenly heated by one or more electrical couplers **704a**, **704b**. In various embodiments, the resistance of each catalytic elements **716** can be adjusted (e.g., by geometrical features, material selection, etc.). For

example, and without limitation, the catalytic elements **716** can be formed from varying densities based on longitudinal location of or within the radial reactor **700**. Radial reactor **700** can have a porous fluid outlet **720** disposed oppositely from the fluid inlet **712**. In various embodiments, the fluid outlet **720** can be configured similarly to the fluid inlet **712**. The fluid inlet **712** and fluid outlet **720** can be generally circular or arcuate in planform shape. In various embodiments, the fluid inlet **712** and fluid outlet **720** can be formed to fully cover the circular end of the electrical couplers **716**. In various embodiments, fluid inlet **712** and fluid outlet **720** can mechanically couple to separators **708** and/or catalytic elements **716** abutting thereto.

[0106] In a first illustrative example of a thermal reactor, the thermal reactor can comprise a reaction module comprising: a first electrode; a catalytic element; and a second electrode; wherein the catalytic element is disposed between the first electrode and the second electrode, wherein the catalytic element is resistively heated to a temperature greater than about 1000° C. by an electrical current provided between the first electrode and the second electrode, wherein a fluid flows through the first electrode into the catalytic element; wherein the first and second electrode determine a flow distribution of fluid through the catalytic element. In variations of the first illustrative example, a pressure drop of the fluid through the first electrode is less than a pressure drop of the fluid through the catalytic element.

[0107] In a second illustrative example, the catalytic element of the first illustrative example can comprise a substrate comprising refractory material and a catalytic material disposed on the substrate, wherein the substrate comprises a high surface area.

[0108] In a third illustrative example, the refractory material of the second illustrative example can comprise at least one of: silicon carbide, tungsten, molybdenum, molybdenum disilicide, tungsten carbide, titanium carbide, graphite, molybdenum carbide, zirconium carbide, hafnium carbide, vanadium carbide, niobium carbide, chromium carbide, tantalum carbide, rhenium carbide, rhodium carbide, ruthenium carbide, osmium carbide, or iridium carbide.

[0109] In a fourth illustrative example, the first electrode and the second electrode of the first, second, or third illustrative example can comprise at least one of: silicon carbide, tungsten, molybdenum, molybdenum disilicide, tungsten carbide, titanium carbide, graphite, molybdenum carbide, zirconium carbide, hafnium carbide, vanadium carbide, niobium carbide, chromium carbide, tantalum carbide, rhenium carbide, rhodium carbide, ruthenium carbide, osmium carbide, or iridium carbide.

[0110] In a fifth illustrative example, the substrate of any of the second, third, or fourth illustrative examples can comprise a foam.

[0111] In a sixth illustrative example, the first electrode and the second electrode of any of the first, second, third, fourth, or fifth illustrative examples are mechanically connected to the catalytic element such that electrical current flows between the first electrode and the second electrode through the catalytic element.

[0112] In a seventh illustrative example, the first electrode and the second electrode of any of the first, second, third, fourth, fifth, or sixth illustrative examples are adhered to the catalytic element using an adhesive, wherein electrical current flows between the first electrode and the second electrode through the adhesive and the catalytic element. In a variation of the seventh illustrative example, the first electrode and the second electrode are connected to the catalytic element by chemical bonds (e.g., the first electrode, the second electrode, and the catalytic element are fabricated as a single object).

[0113] In an eighth illustrative example, the first electrode of any of the first, second, third, fourth, fifth, sixth, or seventh illustrative examples defines through-holes, wherein the fluid flows through the through-holes.

[0114] In a ninth illustrative example, the first electrode of any of the first, second, third, fourth, fifth, sixth, or seventh illustrative examples comprises a mesh, wherein the fluid flows through gaps in the mesh.

[0115] In a tenth illustrative example, the first electrode, the catalytic element, and the second electrode of any of the first, second, third, fourth, fifth, sixth, seventh, eighth, or ninth illustrative examples each have approximately the same cross-sectional size and the same cross-sectional shape.

[0116] In an eleventh illustrative example, the fluid of any of the first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, or tenth illustrative examples comprises hydrogen gas, carbon monoxide, carbon dioxide, and water vapor.

[0117] In a twelfth illustrative example, the thermal reactor of any of the first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, tenth, or eleventh illustrative examples, further comprises a second reaction module, wherein the second reaction module is arranged in a parallel flow stream to the reaction module.

[0118] In a thirteenth illustrative example, the second electrode any of the first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, tenth, eleventh, or twelfth illustrative examples is arranged upstream of the catalytic element relative to the fluid flow.

[0119] In a fourteenth illustrative example, the reaction module (or catalytic element thereof) any of the first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, tenth, eleventh, twelfth, or thirteenth illustrative examples comprises a U-shape or a V-shape.

[0120] In a fifteenth illustrative example, a method for operating a thermal reactor comprises flowing a fluid over or through a catalytic element; and heating the catalytic element to at least a threshold temperature using electrically resistive heating; wherein the fluid flows through void spaces defined by an electrode prior to flowing over or through the catalytic element, wherein the electrode provides electrical current used to resistively heat the catalytic element to the catalytic element.

[0121] In a sixteenth illustrative example, the catalytic element of the fifteenth illustrative example comprises a substrate comprising refractory material and a catalytic material disposed on the substrate, wherein the substrate comprises a high surface area.

[0122] In a seventeenth illustrative example, the refractory material of the sixteenth illustrative example comprises at least one of: silicon carbide, tungsten, molybdenum, molybdenum disilicide, tungsten carbide, titanium carbide, graphite, molybdenum carbide, zirconium carbide, hafnium carbide, vanadium carbide, niobium carbide, chromium carbide, tantalum carbide, rhenium carbide, rhodium carbide, ruthenium carbide, osmium carbide, or iridium carbide.

[0123] In an eighteenth illustrative example, the first electrode and the second electrode of the fifteenth, sixteenth, or seventeenth illustrative example comprises at least one of: silicon carbide, tungsten, molybdenum, molybdenum disilicide, tungsten carbide, titanium carbide, graphite, molybdenum carbide, zirconium carbide, hafnium carbide, vanadium carbide, niobium carbide, chromium carbide, tantalum carbide, rhenium carbide, rhodium carbide, ruthenium carbide, osmium carbide, or iridium carbide.

[0124] In a nineteenth illustrative example, the substrate of any of the sixteenth, seventeenth, or eighteenth illustrative example comprises a foam.

[0125] In a twentieth illustrative example, the fluid of any of the fifteenth, sixteenth, seventeenth, eighteenth, or nineteenth illustrative example comprises hydrogen gas, carbon monoxide, carbon dioxide, and water vapor.

[0126] In a twenty-first illustrative example, either (independently or together) the catalytic element of or an electrical coupler of (singular or plurality) any of illustrative examples 1-14 can be used (e.g., for applications in addition to or as an alternative to thermal reactors such as air or other fluid purifiers, carbon capture, electrolysis, electroplating, etc.).

[0127] In a twenty-second illustrative example, an electrical coupler (e.g., for a resistively heated reactor system, the electrical coupler comprising: an electrical conductor having a first end and a second end, defining a thickness therebetween, each of the first end and the second end having a first interface in the first end and a second interface in the second end; and at least one opening

extending from the first end to the second end, the at least one opening configured to allow a fluid to flow through the electrical coupler.

[0128] In a twenty-third illustrative example, a resistively heated reactor comprising: a proximal end having a planar porous gas inlet; at least one catalytic element having a first end and a second end with a sidewall extending therebetween; a first electrical contactor having a first end and a second end, defining a contactor sidewall therebetween, the contactor sidewall configured to circumscribe a first portion of the sidewall of the at least one catalytic element, a second electrical contactor having a first end and a second end, defining an contactor sidewall therebetween, the contactor sidewall configured to circumscribe a second portion of the sidewall of the at least one catalytic element, the second electrical contactor laterally spaced from the first electrical contactor; a first separator disposed between the first and second electrical contactors circumscribing the at least one catalytic element; a second separator disposed between the first and second contactors circumscribing the at least one catalytic element, the second separator spaced oppositely from the first separator; and a distal end having a planar porous gas outlet.

[0129] In a twenty-fourth illustrative example, the resistively heated reactor of the twenty-third illustrative example, further comprising: a plurality of catalytic elements axially aligned from the proximal end to the distal end; a corresponding plurality of first and second electrical contactors circumscribing each catalytic element, wherein the first and second electrical contactors are separated by the first and the second separator, the first and second separator extending from the proximal end to the distal end, circumscribing a portion of each catalytic element.

[0130] In a twenty-fifth illustrative example, a resistively heated reactor comprising: a pressure vessel having a first vessel end and a second vessel end with a sidewall circumscribing each and extending therebetween; a first electrical coupler disposed at the first vessel end, the first electrical coupler having a first electrical conductor having a first end and a second end, defining a thickness therebetween, at least one opening extending from the first end to the second end, the at least one opening configured to allow a fluid to flow therethrough into the pressure vessel, a second electrical coupler disposed at the second vessel end, the second electrical coupler having a second electrical conductor having a third end and a fourth end, defining a thickness therebetween, at least one opening extending from the third end to the fourth end, the at least one opening configured to allow a fluid to flow from the pressure vessel through the second electrical coupler; a catalytic element disposed within the pressure vessel, the catalytic element in contact with the first electrical coupler and the second electrical coupler.

[0131] In a twenty-sixth illustrative example, the electrical coupler of any of the twenty-second through twenty-fifth illustrative example, wherein the electrical conductor is formed from a refractory metal.

[0132] In a twenty-seventh illustrative example, the electrical coupler the twenty-sixth illustrative example, wherein the refractory metal is at least one of tungsten, molybdenum, niobium, tantalum and/or an alloy thereof.

[0133] In a twenty-eighth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through twenty-seventh illustrative examples, wherein the conductive material is a metal.

[0134] In a twenty-ninth illustrative example, the electrical coupler or thermal reactor of the twenty-eighth illustrative example, wherein the metal is at least one of nickel, copper, gold, titanium, aluminum, or stainless steel.

[0135] In a thirtieth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through twenty-ninth illustrative examples, wherein the electrical conductor is a conductive ceramic.

[0136] In a thirty-first illustrative example, the electrical coupler or thermal reactor of thirtieth illustrative example, wherein the conductive ceramic is at least one of silicon carbide, tungsten carbide, niobium carbides and/or titanium carbide.

[0137] In a thirty-second illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through thirty-first illustrative examples, wherein the electrical conductor is formed from a composite material.

[0138] In a thirty-third illustrative example, the electrical coupler or thermal reactor of the thirty-second illustrative examples, wherein the composite material is a cermet.

[0139] In a thirty-fourth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through thirty-third illustrative examples, wherein the electrical coupler is formed from a single continuous body.

[0140] In a thirty-fifth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through thirty-fourth illustrative examples, wherein the at least one opening is a borehole formed normally to the first end and the second end, the borehole having a diameter of approximately 0.5 to 10 centimeters.

[0141] In a thirty-sixth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through thirty-fourth illustrative examples, wherein the at least one opening is a plurality of openings formed from the first end to the second end, each of the plurality of openings having a diameter of approximately 0.01 to 5 centimeters.

[0142] In a thirty-seventh illustrative example, the electrical coupler or thermal reactor of the thirty-sixth illustrative example, wherein the plurality of openings are formed in ring disposed concentrically with a normal axis extending from a center of the first end to a center of the second end.

[0143] In a thirty-eighth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through thirty-fourth illustrative examples, wherein the at least one opening is a plurality of channels formed in the electrical coupler, the electrical coupler formed as a metallic foam having a plurality of channels therethrough.

[0144] In a thirty-ninth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through thirty-eighth illustrative examples, further comprising at least one electrical interface.

[0145] In a fortieth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through thirty-ninth illustrative examples, further comprising at least one fastener interface.

[0146] In a forty-first illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through fortieth illustrative examples, wherein the first end and the second end each comprise a circular planform shape.

[0147] In a forty-second illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through forty-first illustrative examples, wherein the electrical coupler is formed from a single continuous body.

[0148] In a forty-third illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through forty-second illustrative examples, wherein the pressure vessel forms a seal with each of the first and the second electrical couplers.

[0149] In a forty-fourth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through forty-third illustrative examples, wherein each of the first and the second electrical couplers have a cylindrical shape formed concentrically with the pressure vessel.

[0150] In a forty-fifth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through forty-fourth illustrative examples, wherein at least one of the first and the second electrical couplers have at least one electrical interface.

[0151] In a forty-sixth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through forty-fifth illustrative examples, wherein at least one of the first and the second electrical couplers have at least one fastener interface.

[0152] In a forty-seventh illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through forty-sixth illustrative examples, wherein the first and the second electrical

couplers are coupled via at least one threaded rod extending therebetween and parallel to the pressure vessel.

[0153] In a forty-eighth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through forty-seventh illustrative examples, wherein the catalytic element is contacted by the first electrical coupler by a set screw within a threaded hole of the first electrical coupler, the set screw configured to compress a compressive element against the catalytic element.

[0154] In a forty-ninth illustrative example, the electrical coupler or thermal reactor of forty-eighth illustrative example, wherein the compressive element is a washer.

[0155] In a fiftieth illustrative example, the electrical coupler or thermal reactor of forty-ninth illustrative example, wherein the washer is a Belleville washer or disc spring.

[0156] In a fifty-first illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through forty-eighth illustrative examples, wherein the compressive element is a metal foam.

[0157] In a fifty-second illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through fifty-first illustrative examples, further comprising a thermocouple.

[0158] In a fifty-third illustrative example, a thermal reactor comprising two or more resistively heated reactors (e.g., of any of the twenty-second through fifty-second illustrative examples) fluidly coupled in series, wherein a fluid exiting a first reactor enters a second reactor.

[0159] In a fifty-fourth illustrative example, the electrical coupler or thermal reactor of the fifty-third illustrative example, wherein each of the resistively heated reactors have a common pressure vessel extending therebetween.

[0160] In a fifty-fifth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through fifty-fourth illustrative examples, wherein the first and/or the second electrical coupler is formed from a single continuous body.

[0161] In a fifty-sixth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through forty-fifth or fifty-second through fifty-fifth illustrative examples, wherein the catalytic element and at least one of the first electrical coupler and the second electrical coupler are integrally formed.

[0162] In a fifty-seventh illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through fifty-fifth illustrative examples, wherein at least one of the electrical couplers is formed from a bonded assembly.

[0163] In a fifty-eighth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through fifty-seventh illustrative examples, wherein the pressure vessel is cylindrical.

[0164] In a fifty-ninth illustrative example, the electrical coupler or thermal reactor of any of the twenty-second through fifty-seventh illustrative examples, wherein the pressure vessel is prismatic.

[0165] While the disclosed subject matter is described herein in terms of certain preferred embodiments, those skilled in the art will recognize that various modifications and improvements may be made to the disclosed subject matter without departing from the scope thereof. Moreover, although individual features of one embodiment of the disclosed subject matter may be discussed herein or shown in the drawings of the one embodiment and not in other embodiments, it should be apparent that individual features of one embodiment may be combined with one or more features of another embodiment or features from a plurality of embodiments.

[0166] In addition to the specific embodiments claimed below, the disclosed subject matter is also directed to other embodiments having any other possible combination of the dependent features claimed below and those disclosed above. As such, the particular features presented in the dependent claims and disclosed above can be combined with each other in other manners within the scope of the disclosed subject matter such that the disclosed subject matter should be recognized as also specifically directed to other embodiments having any other possible combinations. Thus, the foregoing description of specific embodiments of the disclosed subject

matter has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosed subject matter to those embodiments disclosed.

[0167] It will be apparent to those skilled in the art that various modifications and variations can be made in the method and system of the disclosed subject matter without departing from the spirit or scope of the disclosed subject matter. Thus, it is intended that the disclosed subject matter include modifications and variations that are within the scope of the appended claims and their equivalents.

[0168] Embodiments of the system and/or method can include every combination and permutation of the various system components and the various method processes, wherein one or more instances of the method and/or processes described herein can be performed asynchronously (e.g., sequentially), contemporaneously (e.g., concurrently, in parallel, etc.), or in any other suitable order by and/or using one or more instances of the systems, elements, and/or entities described herein. Components and/or processes of the preceding system and/or method can be used with, in addition to, in lieu of, or otherwise integrated with all or a portion of the systems and/or methods disclosed in the applications mentioned above, each of which are incorporated in their entirety by this reference.

[0169] As used herein, “substantially” or other words of approximation (e.g., “about,” “approximately,” etc.) can be within a predetermined error threshold or tolerance of a metric, component, or other reference (e.g., within 0.001%, 0.01%, 0.1%, 1%, 5%, 10%, 20%, 30% of a reference), or be otherwise interpreted.

[0170] As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the preferred embodiments of the invention without departing from the scope of this invention defined in the following claims.

Claims

1. A thermal reactor comprising: a catalytic reactor comprising: a first electrode; a second electrode; and a catalytic element; wherein the first electrode, second electrode, and catalytic element comprise substantially the same cross-sectional perimeter, wherein the catalytic reactor is configured such that a current between the first electrode and the second electrode resistively heats the catalytic element; and a preheating element comprising: a first heating element electrode; a second heating element electrode; and a porous heating material; wherein the preheating element is configured such that a current between the first heating element electrode and the second heating element electrode resistively heats the porous heating material; wherein the thermal reactor is configured to allow fluid to sequentially flow through the preheating element and then the catalytic reactor.
2. The thermal reactor of claim 1, wherein a specific surface area of the catalytic element and a specific surface area of the porous heating element are between 10 m.²/g and 1000 m.²/g.
3. The thermal reactor of claim 1, wherein the catalytic element and the porous heating element each comprise at least one of: silicon carbide, tungsten, molybdenum, tungsten carbide, titanium carbide, graphite, molybdenum carbide, zirconium carbide, hafnium carbide, vanadium carbide, niobium carbide, chromium carbide, tantalum carbide, rhenium carbide, rhodium carbide, ruthenium carbide, osmium carbide, or iridium carbide.
4. The thermal reactor of claim 1, wherein the catalytic element and the porous heating element each comprise a foam made from an electrically semiconductive refractory material.
5. The thermal reactor of claim 4, wherein the foam comprises a porosity greater than 5%.
6. A reactor system comprising a plurality of thermal reactors, wherein each thermal reactor of the plurality is a thermal reactor as recited in claim 1, wherein each thermal reactor defines a flow path, wherein each of the flow paths of the reactor system are arranged in parallel.
7. A reactor system comprising a set of thermal reactors, wherein each thermal reactor of the set is the thermal reactor of claim 1, wherein each thermal reactor defines a flow path, wherein each of

the flow paths of the reactor system are arranged sequentially.

8. A method for operating a thermal reactor comprising: flowing a fluid over or through a catalytic element, wherein the catalytic element is disposed between a first electrode and second electrode; and heating the catalytic element by passing a current between the first and second electrode; wherein a perimeter of the catalytic element is substantially identical to a perimeter of the first electrode at a connection interface, wherein a perimeter of the catalytic element is substantially identical to a perimeter of the second electrode at a second connection interface.

9. The method of claim 8, wherein the catalytic element comprises a substrate and a catalytic material disposed on the substrate, wherein the substrate is a foam with a specific surface area that is between 10 m.^{sup.2} per gram and 1000 m.^{sup.2} per gram, wherein the substrate comprises a conductive refractory material.

10. The method of claim 9, wherein the catalytic material comprises at least one of: platinum, palladium, gold, rhodium, ruthenium, copper, nickel, rhenium, cobalt, iron, molybdenum, combinations thereof, oxides thereof, or phosphides thereof; wherein the catalyst comprises a material that depends on the fluid.

11. The method of claim 8, wherein the fluid comprises H._{sub.2} and CO._{sub.2}.

12. The method of claim 11, wherein the H._{sub.2} and CO._{sub.2} reacts to form H._{sub.2}O and CO in the thermal reactor.

13. The method of claim 8, wherein, in the thermal reactor, the fluid undergoes a reverse gas water shift reaction, a steam methane reforming process, a dry methane reforming process, a Haber process, or a Kvaerner process.

14. A thermal reactor comprising: a porous catalytic element comprising a proximal face and a distal face; a first electrode, wherein a face of the first electrode is in direct face-to-face contact with greater than 50% of the area of the proximal face of the porous catalytic element; a second electrode, wherein a face of the second electrode is in direct face-to-face contact with greater than 50% of the area of the distal face of the porous catalytic element; a flow path configured to allow fluid to flow through the porous catalytic element; wherein the thermal reactor is configured such that a current flows in a net direction between the first electrode and the second electrode to resistively heat the porous catalytic element; wherein the flow path defines an average flow path vector that is nonparallel with the net direction of the current.

15. The thermal reactor of claim 14, wherein the face of the first electrode and the proximal face of the porous catalytic element have substantially the same cross-sectional perimeter.

16. The thermal reactor of claim 15, wherein the face of the second electrode and the distal face of the porous catalytic element have substantially the same cross-sectional perimeter.

17. The thermal reactor of claim 14, wherein the porous catalytic element comprises a substrate comprising a surface area between 10 m.^{sup.2} per gram and 1000 m.^{sup.2} per gram; wherein the substrate comprises at least one of: silicon carbide, tungsten, molybdenum, tungsten carbide, titanium carbide, graphite, molybdenum carbide, zirconium carbide, hafnium carbide, vanadium carbide, niobium carbide, chromium carbide, tantalum carbide, rhenium carbide, rhodium carbide, ruthenium carbide, osmium carbide, or iridium carbide.

18. The thermal reactor of claim 14, wherein the porous catalytic element comprises a foam comprising a refractory material with a porosity over 5%.

19. The thermal reactor of claim 14, wherein the porous catalytic element is U-shaped or V-shaped.

20. The thermal reactor of claim 14, wherein the flow path, first electrode, and second electrode are further configured to allow the fluid to flow through at least one of the first electrode and the second electrode.
